Fuzzy Logic

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Fuzzy Learning

Fuzzy learning is a computational approach that utilizes fuzzy logic to handle uncertainty and imprecision in data.

This method is particularly useful in scenarios where traditional binary logic (true/false) is insufficient for making decisions based on vague or ambiguous information

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Fuzzy Logic

Fuzzy logic extends classical logic by allowing for degrees of truth rather than just binary true or false values.

It operates on the premise that many real-world situations cannot be adequately described using strict binary categories.

Introduced by Lotfi Zadeh in the 1960s, fuzzy logic provides a framework for reasoning that mimics human decision-making processes.

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Classical vs Fuzzy logic

Classical logic, also known as Boolean logic, is based on binary values: true or false, 1 or 0. It follows the principles of Aristotle's logic, where statements are either true or false with no middle ground.

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- Classical logic of Aristotle: Law of Bivalence "Every proposition is either True or False(no middle)"

- Jan Lukasiewicz proposed three-valued logic : True, False and Possible

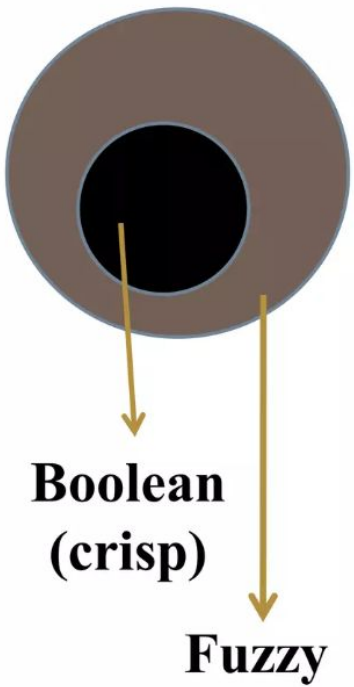
- Finally Lofti Zadeh published his paper on fuzzy logic-a part of set theory that operated over the range [0.0-1.0]

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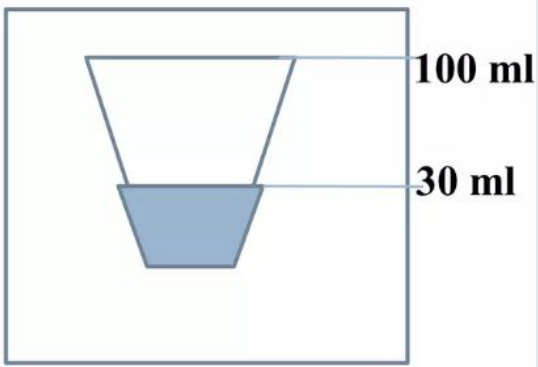
Fuzzy Logic

- Fuzzy logic is a superset of Boolean (conventional) logic that handles the concept of partial truth, which is truth values between "completely true" and "completely false".
- Fuzzy logic is **multivalued.** It deals with **degrees of membership** and **degrees of truth.**
- Fuzzy logic uses the continuum of logical values between 0 (completely false) and 1 (completely true).



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For example, let a 100 ml glass contain 30 ml of water. Then we may consider two concepts: Empty and Full.



In boolean logic there are two options for answer i.e. either the glass is half full or glass is half empty.

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In fuzzy concept one might define the glass as being 0.7 empty and 0.3 full.

Fuzzy sets

- It has an ability to classify elements into a continuous set using the concept of degree of membership.
- Fuzzy set is defined as a set whose elements have degrees of membership.
- The characteristics function or membership function not only gives 0 or 1 but can also give values between 0 and 1.
- Value 0-> non-membership
- Value 1->complete membership
- Value between 0 and 1-> degree of membership

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Crisp and Fuzzy Set

•Classical set theory enumerates all element using
$A=\{a_1,a_2,a_3,a_4\ldots,a_n\}$

Set A can be represented by Characteristic function

$\mu_a(x)=\{$ 1 if element x belongs to the set A
0 otherwise
$\}$

Example: Consider space X consisting of natural number<=12
Prime={x contained in X | x is prime number={2,3,5,7,11}

•In fuzzy set theory elements have varying degrees of membership
A fuzzy set can be represented by:

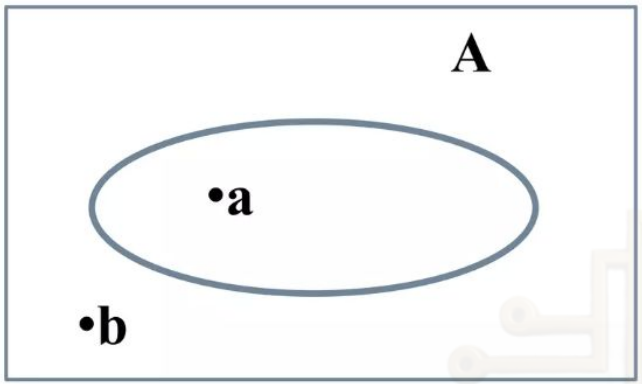
A={{ x, A(x) }}

where, A(x) is the membership grade of a element x in fuzzy set

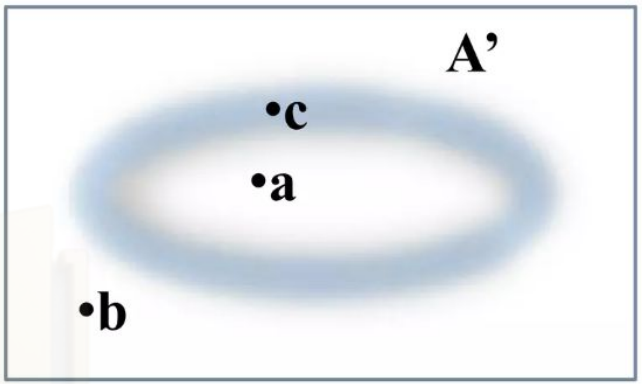
SMALL={{1,1},{2,1},{3,0.9},{4,0.6},{5,0.4},{6,0.3},{7,0.2},{8,0.1},
{9,0},{10,0},{11,0},{12,0}}

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Fuzzy and Crisp set



Crisp set



Fuzzy set

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- a: member of crisp set A
- b: not a member of set A

- a: full member of fuzzy set A'
- b: not a member of set A'
- c:partial member of set A'

Fuzzy and Crisp set

Crisp set

Name	Age	Degree of membership
Sally	5	0
Jenny	18	0
Christen	25	1

Fuzzy set

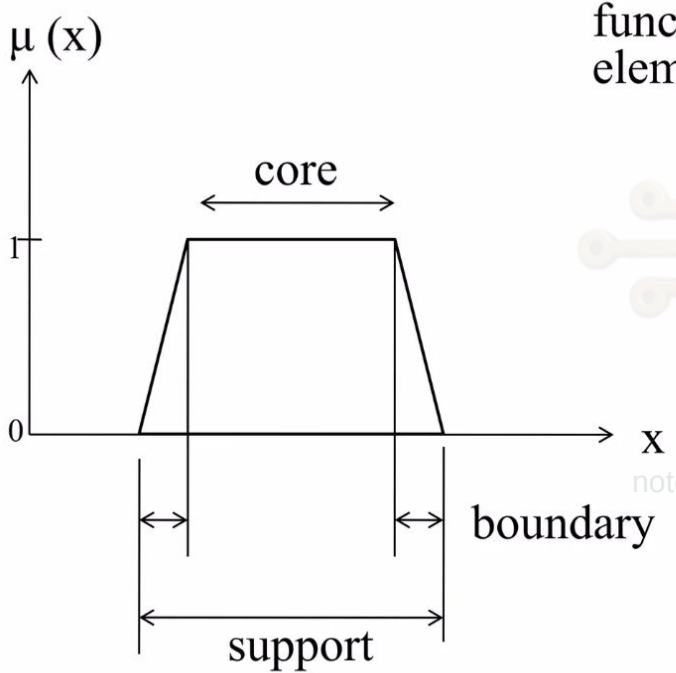
Name	Age	Degree of membership
Sally	5	0
Jenny	18	0.75
Christen	25	1





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Features of member function



A membership function is a mathematical function which defines the degree of an element's membership in a fuzzy set.

- Core: region characterized by full membership in set A' i.e. $\mu(x)=1$.
- Support: region characterized by nonzero membership in set A' i.e. $\mu(x) > 0$.
- Boundary: region characterized by partial membership in set A' i.e. $0 < \mu(x) < 1$

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Membership Function

adult(x)= { 0, if age(x) < 16years

(age(x)-16years)/4, if 16years < = age(x)< = 20years,

1, if age(x) > 20years

}



Fuzzy Logic vs Probability

- Both operate over the same numeric range and at first glance both have similar values:0.0 representing false(or non-membership) and 1.0 representing true.

- In terms of probability, the natural language statement would be ”there is an 80% chance that Jane is old.”

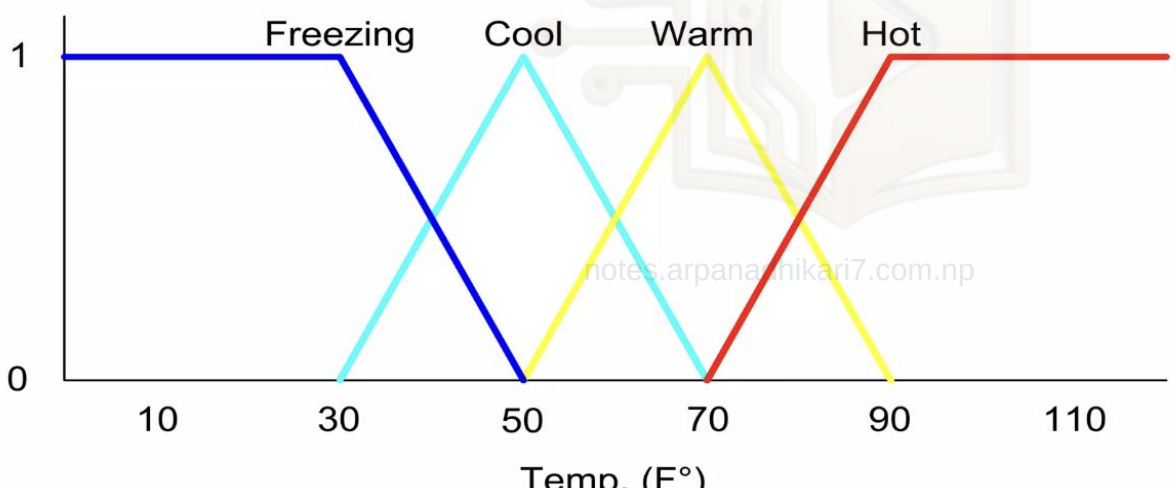
- While the fuzzy terminology corresponds to “Jane’s degree of membership within the set of old people is 0.80.’

- Fuzzy logic uses truth degrees as a mathematical model of the vagueness phenomenon while probability is a mathematical model of ignorance.

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Fuzzy Linguistic Variable

- Fuzzy Linguistic Variables are used to represent qualities spanning a particular spectrum
- Temp: {**Freezing**, **Cool**, **Warm**, **Hot**}

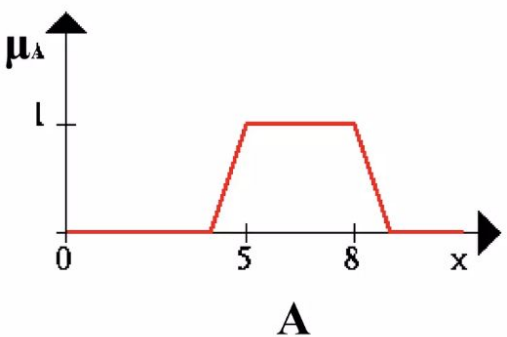


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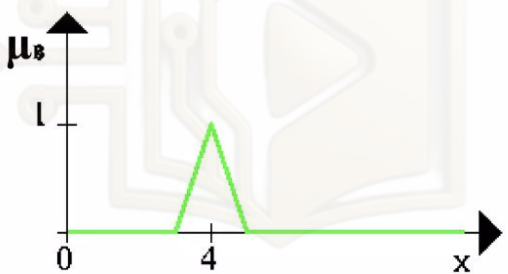
Operations on Fuzzy set

Consider:

A= {1/2 + .5/3 + .3/4 + .2/5} B= {.5/2 + .7/3 + .2/4 + .4/5}



A



B

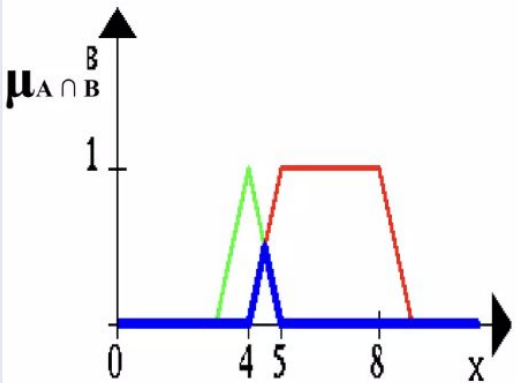
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>Fuzzy set (A)

>Fuzzy set (B)

>Resulting operation of fuzzy sets

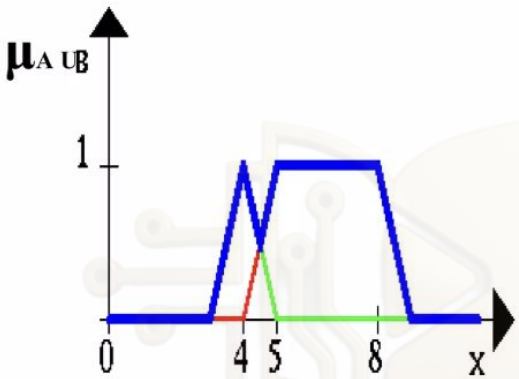
INTERSECTION (A ^ B)



$\mu_{A \cap B} = \min(\mu_A(x), \mu_B(x))$

$\{.5/2 + .5/3 + .2/4 + .2/5\}$

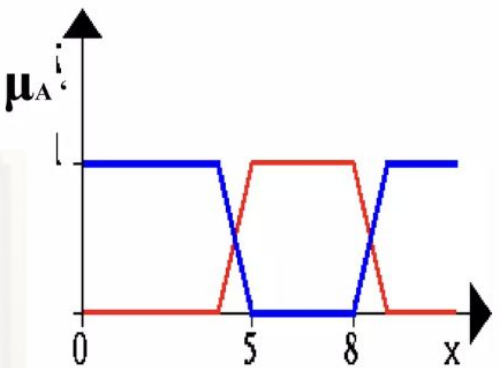
UNION (A v B)



$\mu_{AUB} = \max(\mu_A(x), \mu_B(x))$

$\{1/2 + .7/3 + .3/4 + .4/5\}$

COMPLEMENT (¬A)



$\mu_{A'} = 1 - \mu_A(x)$

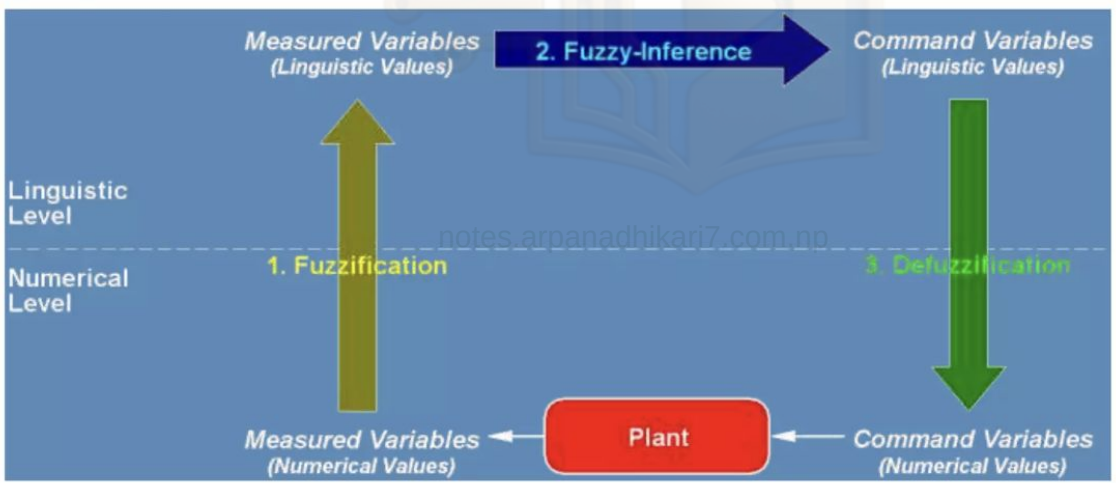
$\{1/1 + 0/2 + .5/3 + .7/4 + .8/5\}$

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Example: Speed Calculation

How fast will I go if it is

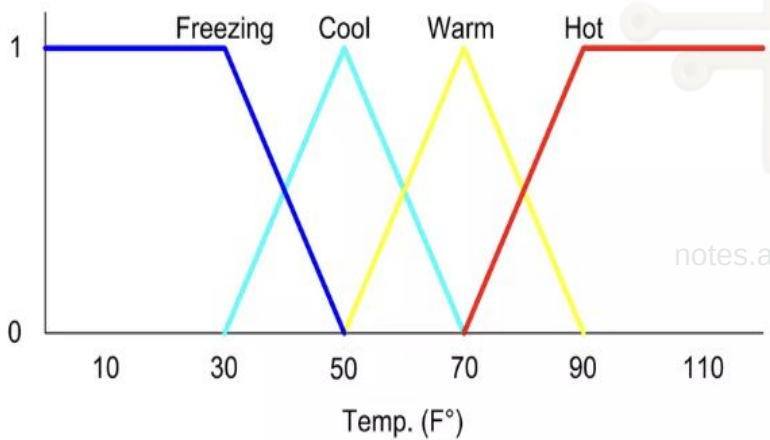
- 65 F
- 25 % Cloud Cover ?



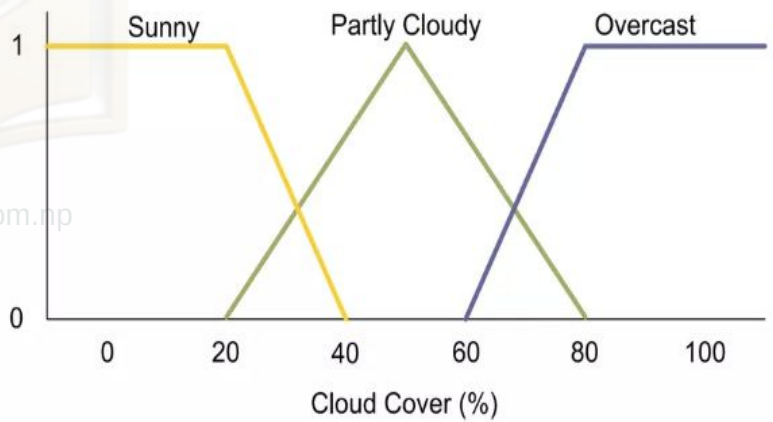
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Input:

Temp: {Freezing, Cool, Warm, Hot}

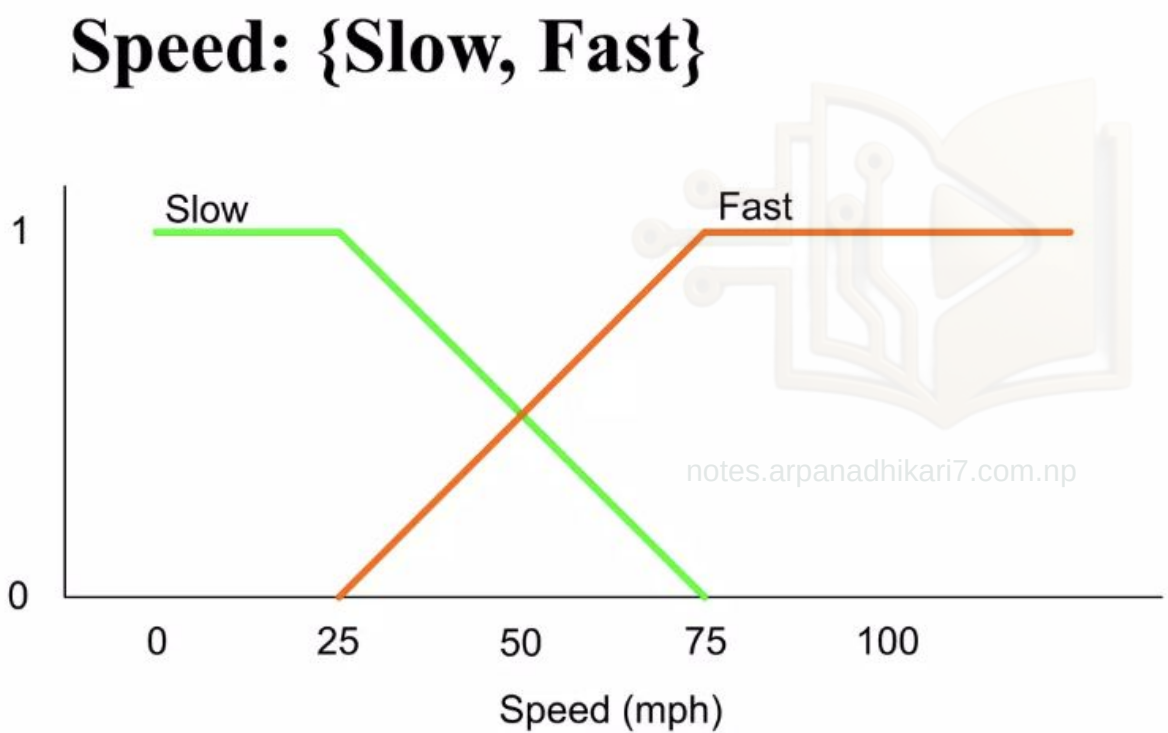


Cover: {Sunny, Partly cloudy, Overcast}



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Output



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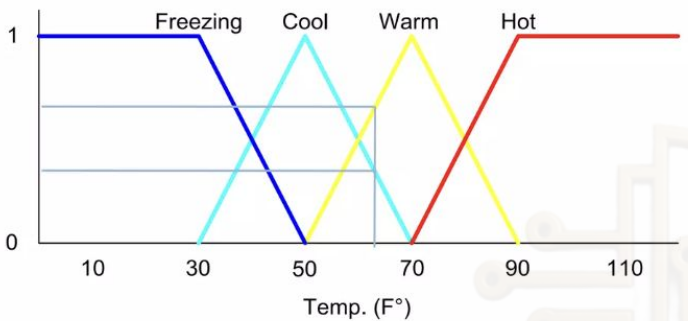
Rules

- If it's Sunny and Warm, drive Fast
  $Sunny(Cover) \wedge Warm(Temp) \Rightarrow Fast(Speed)$
- If it's Cloudy and Cool, drive Slow
  $Cloudy(Cover) \wedge Cool(Temp) \Rightarrow Slow(Speed)$
- Driving Speed is the combination of output of these rules...

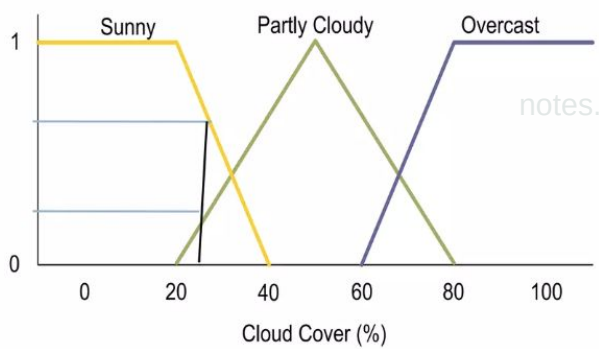
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Fuzzification: Calculate Input Membership Levels

- **65 F ⇒ Cool = 0.4, Warm= 0.7**



- **25% Cover ⇒Sunny = 0.8, Cloudy = 0.2**



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Calculation

- If it's Sunny and Warm, drive Fast

$$\text{Sunny(Cover)} \wedge \text{Warm(Temp)} \Rightarrow \text{Fast(Speed)}$$

$$0.8 \wedge 0.7 = 0.7$$

$$\Rightarrow \textbf{Fast = 0.7}$$

- If it's Cloudy and Cool, drive Slow

$$\text{Cloudy(Cover)} \wedge \text{Cool(Temp)} \Rightarrow \text{Slow(Speed)}$$

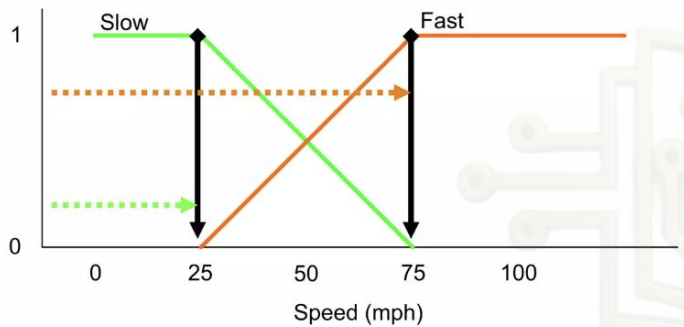
$$0.2 \wedge 0.4 = 0.2$$

$$\Rightarrow \textbf{Slow = 0.2}$$

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Defuzzification: Constructing the output

- Speed is 20% Slow and 70% Fast



- Find centroids: Location where membership is 100%

- Speed = weighted mean
  = (2*25+7*75)/(9)
  = 63.8 mph

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Fuzzy Application

- **Automobile and other vehicle subsystems :** used to control the speed of vehicles, in Anti Braking System.
- **Temperature controllers :** Air conditioners, Refrigerators
- **Cameras :** for auto-focus
- **Home appliances:** Rice cookers , Dishwashers , Washing machines and others

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Drawbacks

- Fuzzy logic is not always accurate. The results are perceived as a guess, so it may not be as widely trusted .
- Requires tuning of membership functions which is difficult to estimate.
- Fuzzy Logic control may not scale well to large or complex problems
- Fuzzy logic can be easily confused with probability theory, and the terms used interchangeably. While they are similar concepts, they do not say the same things.

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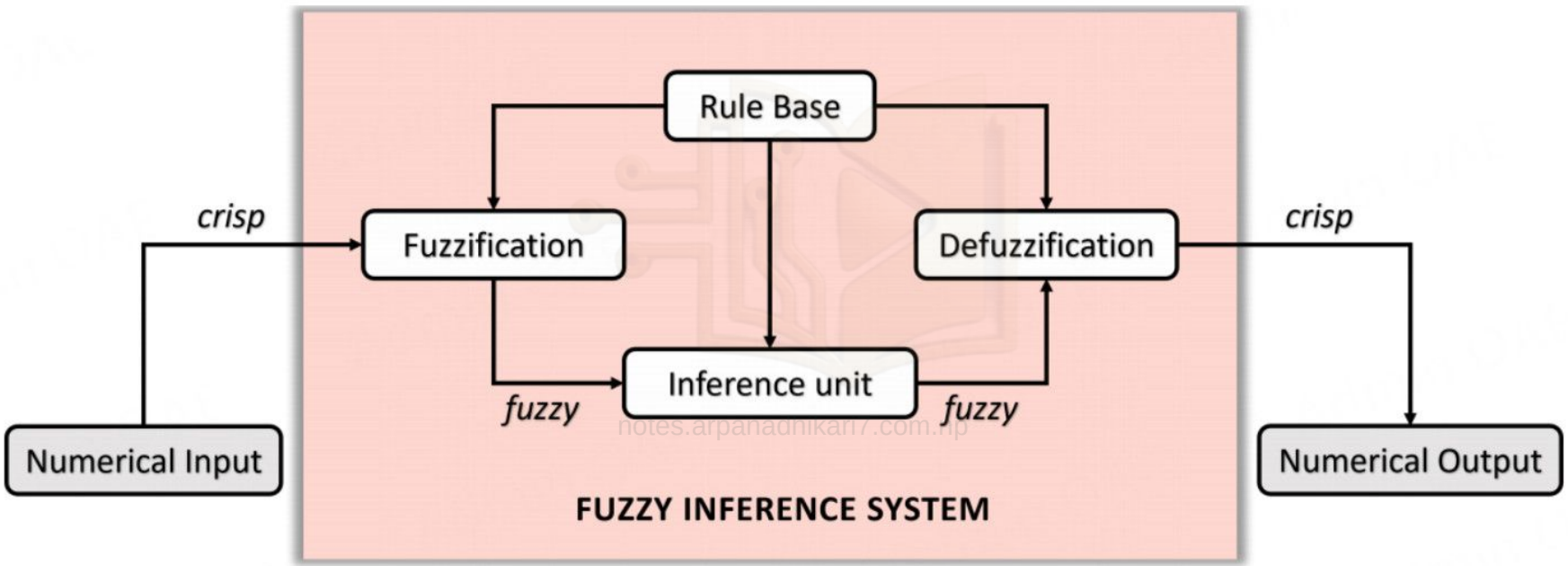
Fuzzy Inference System

A Fuzzy Inference System (FIS) is a framework that uses fuzzy logic to map inputs to outputs.

It processes input data in the form of fuzzy sets, applies rules, and produces a fuzzy output, which is often defuzzified into a crisp value.

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Structure of FIS



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Components of fuzzy inference system

Fuzzification: Converts crisp inputs into fuzzy sets by determining the degree to which they belong to various fuzzy sets.

Knowledge Base: Comprises a set of fuzzy rules and membership functions that define how inputs relate to outputs.

Inference Engine: Applies fuzzy logic rules (IF-THEN) to compute the fuzzy output by combining inputs.

Defuzzification: Converts the fuzzy output back into a crisp value for decision-making.

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Fuzzy Inference Method

Fuzzy inference methods are the ways in which inputs are processed using fuzzy rules to infer outputs in a Fuzzy Inference System (FIS).

These methods determine how fuzzy sets are combined to produce results.

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Fuzzy Inference Method: Mamdani Method

Most commonly used fuzzy inference method.

Employs min-max operations for inference.

Rules are expressed in the form of IF-THEN statements, and the final fuzzy output is defuzzified to get a crisp result.

Example: IF temperature is "high" AND humidity is "low," THEN fan speed is "medium."

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